Euclid's Elements — Book XI
Proposition 7

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJteamLab

If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert's spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid's reductio. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

2 Euclid's Statement

Euclid's Proposition XI.7 states:

> If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD. The claim is that the joining straight line EF lies in the plane of AB and CD.

3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD. The points E and F are arbitrary points on the two lines.

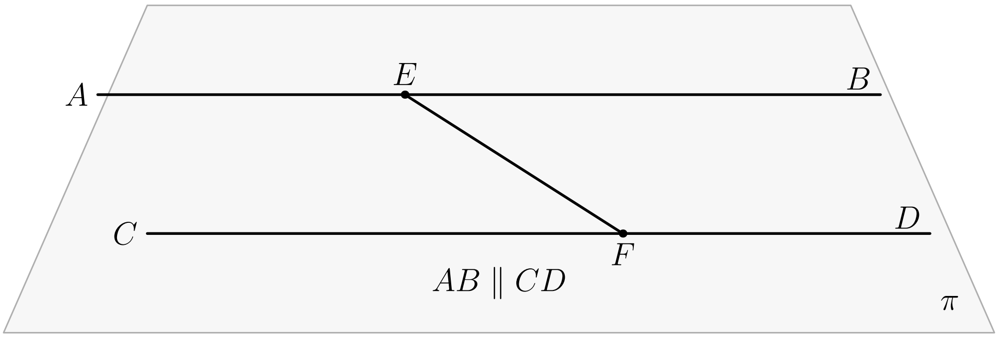

Figure 1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π, with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π. All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F. Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore

$$\begin{gathered}
AB \parallel CD, \quad E \in AB, \quad F \in CD \\
\Downarrow \\
\text{suppose the join } EF \text{ leaves the plane of the parallels} \\
\Downarrow \\
\text{draw a second plane through the alleged elevated join} \\
\Downarrow \\
\text{XI.3} \\
\Downarrow \\
\text{the two planes have a straight-line section through } E, F \\
\Downarrow \\
\text{two distinct straight lines through } E, F \\
\Downarrow \\
\text{impossible} \\
\Downarrow \\
EF \text{ lies in the plane of } AB, CD.
\end{gathered}$$

The only explicit Book XI proposition cited in the proof is XI.3.

5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

$$\text{coplanarity} \qquad \text{and} \qquad \text{disjointness.}$$

XI.7 uses the coplanarity component directly. No metric geometry is needed.

6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_7
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (l m : Geo.Line)
    (E F : Geo.Point)
    (hParallel : HilbertSpaceLinesParallel Geo l m)
    (hEl : H.OnLine E l)
    (hFm : H.OnLine F m) :
    exists sigma : S.Plane,
      HilbertLineInPlane Geo l sigma /\
      HilbertLineInPlane Geo m sigma /\
      exists n : Geo.Line,
```

```
        H.OnLine E n /\
        H.OnLine F n /\
        HilbertLineInPlane Geo n sigma
```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through E and F, together with the proof that `n` lies in `sigma`.

7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel
```

No induced `PlaneGeo` is constructed. There is no planar betweenness, SameRay, SameSide/OppositeSide, angle congruence, right angle, SAS, SSS, or plane–space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

8 Proof Architecture of the Reconstruction

8.1 Step 1: unpack spatial parallelism

From

$$\mathrm{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \qquad m \subset \sigma,$$

together with

$$\mathrm{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general `Hilbert3DInterface` rather than a proposition-local declaration.

8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid's diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m, contradicting

$$\mathrm{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with

$$E \in n, \qquad F \in n.$$

8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n, and both lie in σ. The decisive step is:

```
HilbertSpaceIncidence.line_in_plane
```

which gives

$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert's incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

9 Wyler-Style Flat Reconstruction

XI.7 isolates a particularly reusable generator statement. If $E \in l$ and $F \in m$, then independently of parallelism one has

$$\mathrm{Span}\{E, F\} \subseteq \mathrm{Join}(l, m).$$

This is pure monotonicity of generated flats: the two point generators already belong to the join of the two line carriers.

Parallelism enters only afterwards. It supplies a common plane σ for l, m, identifies their join with the carrier of σ, and guarantees that $E \neq F$. The span of the two points is therefore the carrier of the connector line $n = EF$, and the generic inclusion becomes $n \subseteq \sigma$.

```
E in l, F in m
     |
```

```
       v
Span{E,F} subset Join(l,m)
       |
l || m gives Join(l,m)=carrier(sigma)
       |
E != F gives Span{E,F}=carrier(EF)
       |
       v
EF subset sigma
```

This is a useful example of abstraction at the correct level: the central flat lemma is more general than XI.7, but the public proposition still records the classical parallel-line configuration.

10 Lean Formalization

The proof DAG is extremely short:

$$
\begin{gathered}
\texttt{HilbertSpaceLinesParallel}\ l\ m \\
\Downarrow \\
\exists \sigma, \quad l \subset \sigma, \quad m \subset \sigma, \quad l \cap m = \varnothing \\
\Downarrow \\
E \in \sigma, \quad F \in \sigma \\
\Downarrow \\
E \neq F \\
\Downarrow \\
\texttt{line\_through}\ E\ F \\
\Downarrow \\
n \text{ through } E, F \\
\Downarrow \\
\texttt{HilbertSpaceIncidence.line\_in\_plane} \\
\Downarrow \\
n \subset \sigma \\
\Downarrow \\
\texttt{euclid\_proposition\_11\_7}.
\end{gathered}
$$

There are no proposition-local helper theorems.

11 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase "the straight line joining E and F" is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. The formal spatial incidence interface instead provides the line-in-plane closure principle directly.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

13 Relationship with Book I / Spatial Incidence Infrastructure

Euclid's definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive formal ingredient is the spatial line-in-plane closure principle used by `HilbertSpaceIncidence.line_in_plane`. Thus the proof lands entirely in the spatial incidence layer and nowhere higher in the hierarchy. The source-level identification of this clause inside the classical incidence systems is collected in the Book XI source appendices.

14 Relationship with the Foundations

XI.7 is incidence-only at the geometric level. The proof does not use spatial order, spatial congruence, induced `PlaneGeo`, Hilbert Group IV, or continuity. Once the two parallel lines are unpacked into a common carrier plane, the joining line is obtained by ordinary spatial incidence and the line-closure principle.

This makes XI.7 a useful contrast with XI.6 and XI.8: its content is almost entirely about retaining the correct carrier plane through the construction.

15 Dependency Summary

The classical local chain is

$$\begin{gathered} AB \parallel CD, \quad E \in AB, \quad F \in CD \\ \downarrow \\ \text{contradictory elevated joining line} \\ \downarrow \\ \text{XI.3} \\ \downarrow \\ \text{common section of two planes is a straight line} \\ \downarrow \\ \text{impossible double join through } E, F \\ \downarrow \\ \text{XI.7.} \end{gathered}$$

The actual formal chain is

$$\begin{gathered} \text{Hilbert incidence} \\ \downarrow \\ \text{Hilbert spatial incidence I.6} \\ \downarrow \\ \texttt{HilbertSpaceLinesParallel} \\ \downarrow \\ \text{common carrier plane } \sigma \\ \downarrow \\ E, F \in \sigma, \quad E \neq F \\ \downarrow \\ \texttt{HilbertPlaneIncidence.line\_through} \\ \downarrow \\ \texttt{HilbertSpaceIncidence.line\_in\_plane} \\ \downarrow \\ \texttt{euclid\_proposition\_11\_7}. \end{gathered}$$

16 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The formal reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and `HilbertSpaceIncidence.line_in_plane` closes the line under two distinct points of that plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.